LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 9 Test (Set B): Light – Reflection and Refraction

Chapter / Topic: Light – Reflection and Refraction

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SECTION A

Q1. The radius of curvature of a convex mirror is 32 cm. Its focal length is: **[1]**

Answer: (c) +16 cm. f = R/2, positive for a convex mirror.

Q2. Which of the following is used as a magnifying glass? **[1]**

Answer: (b) Convex lens (object placed between F_1 and O)

Q3. Light travels fastest in: **[1]**

Answer: (d) vacuum

Q4. The magnification produced by a plane mirror is: **[1]**

Answer: (d) +1 (image is the same size and erect)

Q5. When a ray of light passes obliquely from a denser medium into a rarer medium, it: **[1]**

Answer: (c) bends away from the normal

Q6. **Assertion (A):** Convex mirrors are placed at sharp bends on hilly roads. **Reason (R):** A convex mirror gives an erect image and a much wider field of… **[1]**

Answer: (a)

SECTION B

Q7. Define the power of a lens and give its SI unit. Find the power of a convex lens of focal length 40 cm. **[2]**

Answer: The power of a lens is the reciprocal of its focal length in metres (P = 1/f). It shows how strongly the lens converges or diverges light. SI unit: dioptre (D). P = 1/0.40 m = +2.5 D.

- Definition + unit: 1
- P = +2.5 D: 1

OR, answer: It means the ratio of the speed of light in vacuum (air) to the speed of light in diamond is 2.42, so light travels 2.42 times slower in diamond than in vacuum.

- Ratio of speeds: 1
- Light slower in diamond: 1

Q8. Which type of mirror is used (a) in solar furnaces, (b) as a shaving mirror? Give a reason for each. **[2]**

Answer: (a) Concave mirror. It concentrates sunlight at its focus, producing a lot of heat. (b) Concave mirror. When the face is within its focus, it gives an enlarged, erect image.

- Each with reason: 1

SECTION C

Q9. An object 5 cm high is placed 20 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) u = −20 cm, f = −10 cm. 1/v = 1/f − 1/u = −1/10 + 1/20 = −1/20, so v = −20 cm. The image is 20 cm in front of the mirror (at C). [2]

- Formula and signs: 1
- v = −20 cm: 1

(ii) m = −v/u = −(−20)/(−20) = −1, so h′ = −5 cm. The image is real, inverted and the same size as the object. [1]

Q10. A convex lens of focal length 10 cm forms a real image three times the size of the object. Find the object distance and the image distance. **[3]**

Answer: Real image, so m = −3 and v = −3u. Using 1/v − 1/u = 1/f: 1/(−3u) − 1/u = 1/10, which gives −4/(3u) = 1/10, so u = −40/3 ≈ −13.3 cm. v = −3u = +40 cm. The object is 13.3 cm from the lens, and the image is 40 cm on the other side.

- m = −3, v = −3u: 1
- u ≈ −13.3 cm: 1
- v = +40 cm: 1

OR, answer: u = −20 cm, f = −20 cm. 1/v = 1/f + 1/u = −1/20 − 1/20 = −1/10, so v = −10 cm (same side as the object). m = v/u = (−10)/(−20) = +0.5. The image is virtual, erect and half the size of the object.

- v = −10 cm: 1
- m = +0.5: 1
- Nature: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Diagram: a ray parallel to the principal axis reflects through F, and a ray through C reflects back along itself. They meet beyond C. Image: beyond C; enlarged; real and inverted. [3]

- Correct ray diagram: 2
- Position, size and nature: 1

(ii) $P = P_1 + P_2 = +3.5 + (−2.5) = +1.0$ D. f = 1/P = 1 m = 100 cm. The positive power means the combination is converging. [2]

- P = +1 D: 1
- f = 1 m, converging: 1

OR, answer: (a) (1) The angle of incidence equals the angle of reflection. (2) The incident ray, the normal at the point of incidence and the reflected ray all lie in the same plane. (b) Light from the coin passes from water (denser) into air (rarer) and bends away from the normal. Our eyes trace the refracted rays back in straight lines, and they appear to come from a point higher than the coin. So the coin looks raised. This is due to refraction.

- (a) 1 + 1
- (b) refraction 1, bends away 1, appears raised 1

SECTION E

Q12. In an experiment with a rectangular glass slab, a student sent a ray of light into the slab at an angle of incidence of 30°. The angle of refraction… **[4]**

(i) For a given pair of media and a given colour of light, sin i / sin r is a constant (the refractive index of the second medium with respect to the first). [1]

(ii) n = sin i / sin r = 0.500/0.334 ≈ 1.5 [1]

(iii) The angle of refraction increases. The refractive index stays the same (about 1.5), because it depends only on the pair of media (and the colour of light), not on the angle. [2]

- r increases: 1
- n unchanged, with reason: 1